AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# MT132/M132
# Linear Algebra

## Midterm Examination (MTA)
## Spring Semester 2024/2025

## Date: 10 /April/2025

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| **Number of Exam Pages: 3** (including this cover sheet) | **Time Allowed:** 90 Minute |

**Instructions:**

- Total Marks: **30**
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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**Q–1: [4×2 marks] Choose the correct answer:**

**1.1:** If $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ then $(A^{-1})^3 + I_2 =$

a) $I_2$
b) $\mathbf{0}$
c) $-I_2$
d) $-A$
e) None of the above

**1.2** The matrix $\begin{bmatrix} a & 1 \\ 1 & a \end{bmatrix}$ is singular if

a) $a = 0$
b) $a \neq 0$
c) $a = \pm 1$
d) $a \neq \pm 1$
e) None of the above

**1.3:** If $A = BDB^{-1}$ then

a) $A^3 = B^3D^3(B^{-1})^3$
b) $A^3 = BDB^{-1}$
c) $A^3 = BD^3B^{-1}$
d) $A^3 = D^3$
e) None of the above

**1.4:** If $A$ *and* $B$ are symmetric square matrices, then

a) $ABA$ is symmetric
b) $ABA$ is skew-symmetric
c) $AB$ is symmetric
d) $AB$ is skew-symmetric
e) None of the above

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**Q–2: [8 marks] Answer the following questions as True or False (1 mark), Justifying your answer (3 marks):**

**2.1:** If $A$ *and* $B$ are $2 \times 2$ matrices, where $B = 3A$, and $|A| = 2$, then $|B| = 18$ .

**2.2:** for the nonsingular matrix $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$, $\begin{vmatrix} c & d \\ c-a & b-d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}$.

**Each question is worth 7 marks. Answer the following questions:**

**Q–3 [ 4+3 marks]:**

**a)** Let $A = \begin{bmatrix} -1 & 1 & 0 \\ -2 & 1 & 1 \\ 2 & -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 3 \\ 0 & 2 & 0 \end{bmatrix}$.

i- Find the inverse of the coefficient matrix $A$, $A^{-1}$.

ii- Find the matrix $X$ such that $(I + X)A = B$.

**b)** Consider the linear system $\begin{cases} x + 2y = a \\ 3x + 6y = 9 \end{cases}$,

For which values of $a$ does the linear system have no solution, a unique solution or infinitely many solutions? Then solve the linear system at $a = 3$.

**Q–4 [4+3 marks]:**

**a)** Let $A^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \\ 0 & 1 & 0 \end{bmatrix}$.

i- Find the determinant of $A$.

ii- Find $|2A^{-1}A^2|$.

iii- Is $A$ invertible? Explain.

iv- Solve the linear system $A^T X = 0$?

**b)** Let $A = \begin{bmatrix} 1 & 2 \\ 0 & a \end{bmatrix}$, find all values of $a$ for which $A^2$ is symmetric?

**End of questions**

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